

Explainable AI For NBA Predictions- Extended

Anonymous Author(s)

Abstract

This paper proposes an explainable artificial intelligence framework for predicting NBA game outcomes using both pre-game statistics and live play-by-play data. The system integrates a Support Vector Machine (SVM) model for pre-game win prediction with an XG-Boost model that updates win probabilities throughout a game using play-by-play snapshots. To improve interpretability, SHAP (SHapley Additive exPlanations) is applied to the pre-game model to identify the statistical factors most responsible for each prediction. The models are trained using NBA data from the 2024–2025 and 2025–2026 seasons obtained through the `nba_api`. Initial evaluation shows that the pre-game SVM model achieves approximately 79% validation accuracy, outperforming a simple baseline that predicts the pre-game favorite. This framework demonstrates how predictive performance and explainability can be combined to produce interpretable sports analytics systems.

Introduction

Predictive analytics has become an important component of modern sports analysis, particularly in basketball, where win-probability models estimate the likelihood that a team will win at different points during a game. These models combine historical game data with machine learning techniques to generate predictions based on factors such as score differential, time remaining, team efficiency metrics, and recent performance trends. While many recent studies have improved the predictive accuracy of sports outcome models, much less attention has been given to how users interpret and understand these predictions.

Many machine learning systems operate as “black boxes,” making it difficult for non-expert users to understand why predictions are generated or why they change during a game. In sports analytics contexts, this lack of interpretability limits the usefulness of predictive models for fans, analysts, and commentators who may want to understand the statistical reasoning behind predictions.

The motivation for this research is to develop an explainable prediction system that combines predictive accuracy with transparent reasoning. Rather than focusing only on

prediction performance, this work investigates how explainable machine learning techniques can help users understand which statistical factors influence predicted outcomes.

This study investigates the following research questions:

RQ1: How accurately can NBA game outcomes be predicted using pre-game team statistics and momentum indicators?

RQ2: How can live play-by-play data be used to dynamically update win probability predictions during a game?

RQ3: How can explainable AI techniques help users understand the factors driving NBA game predictions?

Paper Type

This project is an **experimental systems paper**. The contribution is an explainable machine learning framework that integrates pre-game prediction, live in-game prediction, and feature-level explanations for NBA win probability estimation.

Dataset

We use NBA game data accessed via `nba_api`, a Python client for official NBA statistics endpoints. The dataset includes team-level game logs and play-by-play data from the 2024–2025 and 2025–2026 NBA seasons. While the data is initially retrieved through `nba_api`, the system continuously downloads and stores season statistics locally as CSV files. This design ensures that the prediction pipeline does not depend on real-time API availability and allows the full dataset used for training and evaluation to be preserved. Storing the data locally also improves reproducibility, since future researchers can rerun the experiments without relying on external API endpoints.

Data sources. Fields include team shooting efficiency metrics, scoring statistics, rebounds, assists, turnovers, and play-by-play events such as score changes and time remaining.

Preprocessing pipeline.

- Historical NBA regular-season game logs for two seasons
- Construction of game-level feature vectors for pre-game prediction
- Construction of play-by-play game-state snapshots
- Feature standardization and dataset cleaning

- Train/test splits that prevent leakage across games

To emphasize recent performance, statistics from the current season are given greater weight than the previous season. The system blends statistics using an 85% weight for the current season and a 15% weight for the previous season.

Ethical considerations. The dataset contains only publicly available sports statistics and does not include any personal or sensitive information.

Literature Review

Recent work in AI-driven sports analytics shows rapid progress in predictive accuracy but limited focus on interpretability. Our prior framework (Abdulla Ali 2026) integrates pre-game and in-game NBA win prediction with SHAP-based explanations to improve model transparency.

Graph-enhanced models extend traditional tabular prediction by incorporating relationships between teams in a league structure. Zhao et al. (Zhao, Du, and Tan 2023) demonstrate that combining graph convolutional networks with ensemble models improves predictive stability.

Explainable boosting methods have also been applied to sports analytics. Ouyang et al. (Ouyang et al. 2024) integrate XGBoost models with SHAP explanations and show that feature importance varies throughout different phases of a basketball game.

Broader surveys of sports outcome prediction highlight the growing use of machine learning models in sports forecasting. Bunker and Susnjak (Bunker and Susnjak 2022) note that while many models achieve high predictive performance, interpretability remains a major challenge for practitioners.

These studies motivate the development of predictive systems that balance accuracy with explainability.

Methodology

The proposed system consists of two main components: a pre-game prediction model and a live in-game prediction model. Together, these models produce win probability estimates that can be updated throughout the course of a basketball game.

Prediction Pipeline

The prediction pipeline begins with a pre-game model that estimates the probability that the home team will win before the game begins. These initial probabilities are then updated during the game using play-by-play data.

For each game, the system constructs a set of statistical features describing the relative strength of the two teams. During the game, additional features describing the current state of play are incorporated to update predictions.

Pre-Game Model

The pre-game prediction model uses a Support Vector Machine (SVM) classifier with a radial basis function (RBF) kernel. Features are standardized using a `StandardScaler` prior to training.

Pre-game features include differences between the home and away teams in several statistical categories:

- Field goal percentage
- Three-point percentage
- Free throw percentage
- Points, rebounds, assists, and turnovers
- Net scoring margin
- Recent win percentage over the last ten games
- Elo-based team strength ratings

Each game is represented as a feature vector consisting of the difference between the home team and away team values for these statistics. The model outputs the probability that the home team will win the game.

The model is trained using an 80/20 train-validation split. Initial experiments show that the SVM model achieves approximately 79% validation accuracy.

Live Game Model

To capture the dynamic nature of basketball games, a second model generates win probability predictions using play-by-play data. Game states are represented as snapshots extracted from play-by-play events.

Each snapshot contains contextual information describing the state of the game, including:

- Score differential
- Time remaining in the game
- Current period
- Recent scoring runs
- Momentum indicators derived from score changes

An XGBoost classifier is trained on these snapshots to estimate the probability that the home team will ultimately win the game given the current game state.

Explainability

To improve interpretability, SHAP (SHapley Additive exPlanations) values are computed for predictions generated by the pre-game SVM model. SHAP values quantify the contribution of each feature to the predicted win probability.

The most influential features are presented as explanations that describe the statistical advantages or disadvantages of each team. These explanations provide insight into why the model favors one team over the other.

Results

The pre-game SVM model achieves approximately 79% validation accuracy when evaluated on held-out data. This performance exceeds a simple baseline approach that predicts the winner based on the pre-game favorite, which historically wins approximately 66% of NBA games (Covers 2024).

Because the live model generates predictions continuously during the game, evaluating its accuracy requires defining specific time points for comparison. The evaluation of the live prediction model will be explored further in future work.

Discussion

The results suggest that combining team statistics, momentum indicators, and Elo ratings can produce accurate predictions for NBA game outcomes. In addition, explainable AI techniques provide insights into which statistical factors drive predictions.

One challenge in evaluating live win-probability models is determining how predictions should be measured across different points in the game. Future work will explore standardized evaluation approaches for play-by-play prediction systems.

Conclusion

This paper presents an explainable AI framework for predicting NBA game outcomes using both pre-game statistics and live play-by-play data. The system combines a Support Vector Machine pre-game model with an XGBoost live prediction model and integrates SHAP explanations to improve interpretability.

Initial results demonstrate strong predictive performance for the pre-game model and highlight the potential for combining predictive modeling with explainable AI techniques in sports analytics systems. Future work will focus on evaluating live predictions and expanding the explainability framework.

References

- Abdulla Ali, Masih Faryadi, Y. S. Z. A.-A. 2026. Explainable ai for nba win prediction.
- Bunker, R., and Susnjak, T. 2022. A review of machine learning methods for sport outcome prediction. *Applied Artificial Intelligence* 36(1).
- Covers. 2024. Nba league betting trends: Season to date. <https://www.covers.com/sport/basketball/nba/league-trends/seasontoday>. Accessed: 2026-03-04.
- Ouyang, Y.; Li, X.; Zhou, W.; Hong, W.; Zheng, W.; Qi, F.; and Peng, L. 2024. Integration of machine learning xgboost and shap models for nba game outcome prediction and quantitative analysis methodology. *PLOS ONE* 19(7):e0307478.
- Zhao, K.; Du, C.; and Tan, G. 2023. Enhancing basketball game outcome prediction through fused graph convolutional networks and random forest algorithm. *Entropy* 25(5):765.